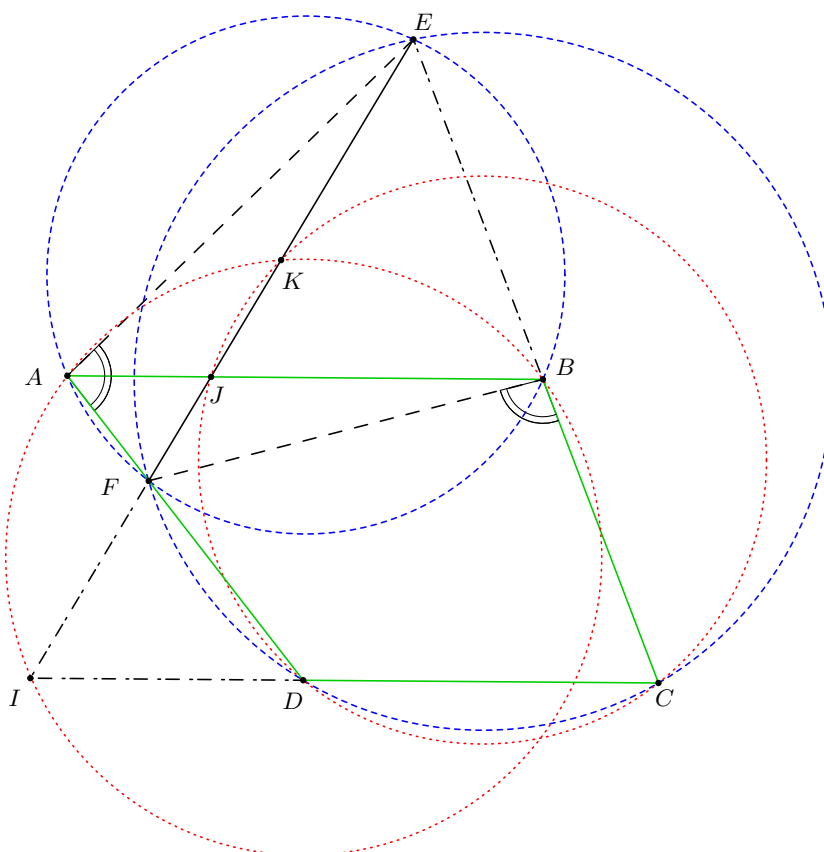


Problem 1. Given trapezoid $ABCD$ with parallel sides AB and CD , assume that there exist points E on line BC outside segment BC , and F inside segment AD , such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF , and by J the point of intersection of AB and EF . Let K be the midpoint of segment EF ; assume that it does not lie on line AB . Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ .



Solution. There are several configurations but the proofs are quite similar. In the following we only focus on the above configuration. Since $\angle EBF = 180^\circ - \angle CBF = 180^\circ - \angle EAF$ by hypothesis, the quadrilateral $AEBF$ is cyclic. Hence $AJ \cdot JB = FJ \cdot JE$. In view of this equality, I belongs to the circumcircle of ABK if and only if $IJ \cdot JK = FJ \cdot JE$. Since

$$IJ = IF + FJ, \quad JE = FE - FJ, \quad \text{and} \quad JK = \frac{1}{2}FE - FJ,$$

we find that I belongs to the circumcircle of ABK if and only if

$$FJ = \frac{IF \cdot FE}{2IF + FE}.$$

Since $AEBF$ is cyclic and $AB \parallel CD$, then $\angle FEC = \angle FAB = 180^\circ - \angle CDF$. Hence $CDFE$ is also cyclic, yielding $ID \cdot IC = IF \cdot IE$. So K belongs to the circumcircle of CDJ if and only if $IJ \cdot IK = IF \cdot IE$. Since

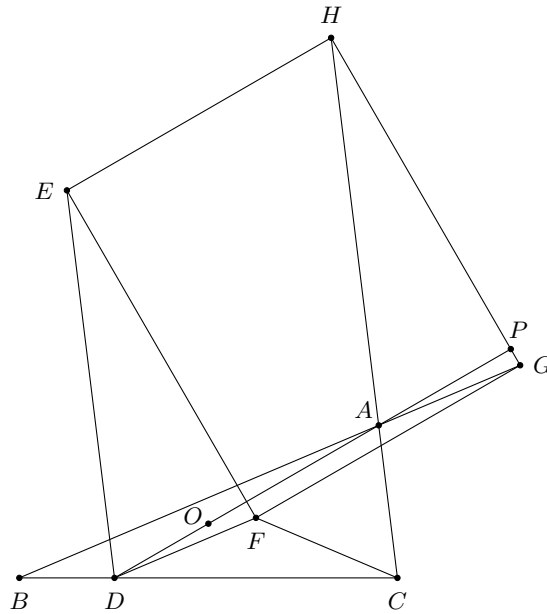
$$IJ = IF + FJ, \quad IK = IF + \frac{1}{2}FE, \quad \text{and} \quad IE = IF + FE,$$

we find that K belongs to the circumcircle of CDJ if and only if

$$FJ = \frac{IF \cdot FE}{2IF + FE}.$$

The result follows. □

Problem 2. Let O be the circumcentre of an acute-angled triangle ABC with $\angle B < \angle C$. The line AO meets the side BC at D . The circumcentres of $\triangle ABD$ and $\triangle ACD$ are E and F , respectively. Extend the sides BA and CA beyond A , and choose on the respective extensions points G and H such that $AG = AC$ and $AH = AB$. Prove that the quadrilateral $EFGH$ is a rectangle if and only if $\angle ACB - \angle ABC = 60^\circ$.



Solution. Let $\angle ABC = \beta$, $\angle ACB = \gamma$, where $\beta < \gamma$. Since $\triangle ABC$ is acute with O its circumcentre, $\angle OAC = 90^\circ - \beta$. Then $\angle ADB = \angle OAC + \angle ACD = 90^\circ - \beta + \gamma > 90^\circ$, so that $\triangle ABD$ is obtuse and $\triangle ACD$ is acute. Hence F is inside $\triangle ACD$ and E is outside $\triangle ABD$; O and E are on different sides of AB .

Because F and O are the circumcentres of $\triangle ACD$ and $\triangle ABC$, we have

$$\angle FAC = 90^\circ - \angle ADC = \gamma - \beta, \quad \angle FDC = 90^\circ - \angle OAC = \beta$$

Hence $FD \parallel AB$; analogously, $ED \parallel AC$. Denote by P the intersection of AD and GH . Then $\angle PAH = \angle OAC = 90^\circ - \beta$. On the other hand $\angle PHA = \angle ABC = \beta$, because $\triangle ABC$ and $\triangle AHG$ are congruent. Thus $\angle APH = 90^\circ$, meaning that $AD \perp GH$. But EF is the perpendicular bisector of AD , so $AD \perp EF$. It follows that $GH \parallel EF$.

So the corresponding sides of $\triangle AHG$ and $\triangle DEF$ are parallel. Then HE and GF either intersect on AD , or they are parallel to AD . The lines EF and GH are perpendicular to AD , so $EFGH$ is a rectangle if and only if the second alternative holds. In view of $DE \parallel AH$ and $DF \parallel AG$, this is the case if and only if the (similar) triangles AHG and DEF are congruent; i.e., if and only if $DF = AG$. Because F is the circumcentre of $\triangle ACD$ and $AG = AC$, the latter is in turn equivalent to $AC = CF = AF$; that is, to ACF being an equilateral triangle.

Since $\triangle ACF$ is isosceles (with base AC), it is equilateral if and only if $\angle FAC = 60^\circ$. And since $\angle FAC = \gamma - \beta$, the claim follows. $\square$